
GeoAgent: Overcoming Step Degradation in Geospatial Vision-Language Models via Tool-Augmented Agentic Reasoning

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Abstract

GitHub Repository: <https://github.com/Ali-Abdullah-1921/DL-project>
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This paper investigates the phenomena of step degradation and ontology mismatch in Vision-Language Models (VLMs) during multi-hop spatial reasoning over remote sensing imagery. To address these critical bottlenecks, we propose the “V2 GeoAgent” architecture: an autonomous ReAct loop built upon Qwen2-VL that integrates Oracle-backed vision tools and introduces Recurrent Visual Attention Masking (RVAM). Furthermore, we implement a rigorous Adversarial Integrity Suite to definitively prove that accuracy gains are driven by genuine spatial logic rather than linguistic hallucination. Evaluated on a mathematically balanced 600-question spatial benchmark, GeoAgent achieves an 84.5% accuracy on strict 4-hop tasks (a +31.5% absolute improvement over the baseline), demonstrating robust, verifiable multi-step reasoning over complex satellite data.

1. Related Work

The development of GeoAgent intersects three primary domains of deep learning: large vision-language models for remote sensing, tool-augmented programming, and visual attention mechanisms.

1.1. Vision-Language Models in Remote Sensing

Recent advancements have adapted generic VLMs, such as LLaVA (Liu et al., 2023), to the remote sensing domain (Kuckreja et al., 2024). A persistent limitation of these models has been the rigid compression of images into low-resolution squares. The introduction of Qwen2-VL (Wang et al., 2024) resolved this bottleneck by implementing Naive Dynamic Resolution and Multimodal Rotary Position Embeddings (M-ROPE). Despite these upgrades, rigorous eval-

uation (Li et al., 2023; Yue et al., 2024) reveals that VLMs still suffer from spatial hallucination when subjected to multi-hop reasoning queries.

1.2. Tool-Augmented Visual Reasoning

To overcome the spatial limitations of frozen LLM weights, frameworks such as ViperGPT (Suris et al., 2023), Visual Programming (VisProg) (Gupta et al., 2023), and ReAct (Yao et al., 2023) allow models to call discrete APIs to solve tasks compositionally. By generating Chain-of-Thought (CoT) reasoning traces (Wei et al., 2022), the LLM acts as an orchestrator. In geospatial contexts, this involves leveraging foundation models like Segment Anything (SAM) (Kirillov et al., 2023; Zhang et al., 2023) and Bitemporal Image Transformers (Chen et al., 2021) rather than relying exclusively on the LLM’s weak internal spatial representations.

1.3. Visual Attention Masking

A critical failure point in multi-hop Visual Question Answering (VQA) is “attention collapse,” wherein a VLM fixates on a primary anchor object and ignores subsequent targets (Hudson & Manning, 2018). Inspired by the Hide-and-Seek architecture (Singh et al., 2017)—which forces networks to discover secondary features by physically masking discriminative regions—we introduce Recurrent Visual Attention Masking (RVAM).

2. Methodology and Results

This project was executed in three progressive phases to establish an unbiased baseline, isolate reasoning bottlenecks, and systematically eliminate failure points through architectural enhancements.

2.1. Task 3: Baseline

To rigorously evaluate spatial reasoning, we developed a strict, unbiased 600-question benchmark using the **xView** dataset (Lam et al., 2018). Questions were programmatically generated to ensure a perfect 50/50 target distribution across 2-Hop, 3-Hop, and 4-Hop spatial queries.

Evaluating the **LLaVA-1.5 Baseline** revealed severe hallucination rates. A “Blind Stress Test” (in which the model was fed a completely black image) yielded positive accuracy, empirically proving the baseline was exploiting linguistic correlations rather than visually grounding its answers.

2.2. Task 4: First Improvement

To mitigate these hallucinations, we introduced a **ReAct** loop utilizing LLaVA-1.5 alongside **YOLOv8-ORB** (Xia et al., 2018) and SAM. While this grounded the model, it introduced a *Compounding Error Penalty*: if the object detector missed a target due to environmental noise, the agent failed the reasoning task regardless of its logical soundness. Furthermore, our initial RVAM implementation utilized solid black boxes to mask objects, which caused destructive interference by obscuring adjacent contextual terrain.

2.3. Task 5: Second Improvement (Final Architecture)

To isolate true reasoning capabilities from predictive detector failures, we developed the final V2 Architecture:

- **Qwen2-VL Transition:** Replaced LLaVA with Qwen2-VL to leverage dynamic resolution scaling and superior instruction-following.
- **Oracle GeoTools:** Replaced YOLO with Oracle tools backed directly by the xView GeoJSON ground truth. This mathematically isolates the VLM’s reasoning logic from perceptual blindness (Suris et al., 2023).
- **Gaussian RVAM:** Replaced solid black boxes with a Heavy Gaussian Blur (`cv2.GaussianBlur`). This effectively scrambles discriminative features while preserving surrounding pixels, forcing the ReAct loop to smoothly re-evaluate the scene.

A core contribution of this phase is the implementation of an **Adversarial Integrity Suite**. We utilized Poisoned Data Tests (inverting ground truth to ensure the regex-based automatic grader was not “leaking” points) and Reasoning Bypass Audits (scanning the internal CoT to flag correct answers achieved without tool calls) to guarantee that reported accuracies reflect genuine System-2 reasoning.

2.4. Final Results and Comparative Analysis

Table 1. Spatial Reasoning Accuracy: Baseline vs. V2 GeoAgent

ARCHITECTURE	2-HOP	3-HOP	4-HOP
LLaVA-1.5 BASELINE	57.0%	59.5%	53.0%
V2 GEOAGENT	89.5%	83.0%	84.5%

As demonstrated in Table 1, the V2 GeoAgent system-

atically dismantles the baseline’s limitations. While the baseline suffers from severe step degradation—dropping to nearly random guessing (53.0%) on 4-hop queries—the GeoAgent maintains highly robust logical grounding, achieving **84.5% accuracy on the same 4-hop tasks (a +31.5% absolute improvement)**.

Reasoning Trace Example (V2):

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Question: Find the Aircraft on the left.
Now move your focus to the right half. Is
there a Cargo Truck located in that right
half?

[LLM]: Thought: I need to locate the planes
on the left, but I shouldn't be distracted
by them when checking the right.
Action: visual_masker Aircraft

[TOOL]: RVAM: Masked 2 'Aircraft' instance(s)
with Gaussian Blur. Attention redirected.

[LLM]: Thought: Now I can clearly scan the
right half for cargo trucks.
Action: spatial_query Cargo Truck right

[TOOL]: CONFIRMED: 1 Cargo Truck(s) present
in RIGHT half.

[LLM]: Final Answer: Yes
    
```

3. Conclusion

In this paper, we proposed GeoAgent to solve step degradation in remote sensing VLMs. By transitioning to a Qwen2-VL ReAct loop utilizing Oracle-backed tools and Gaussian Recurrent Visual Attention Masking (RVAM), we successfully isolated and mathematically grounded the model’s reasoning capabilities. Crucially, the implementation of our Adversarial Integrity Suite guarantees that the 84.5% accuracy achieved on strict 4-hop spatial tasks is a result of genuine, step-wise logical deductions rather than linguistic hallucination.

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